BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Ethical and Social Impacts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why can defragmentation help a magnetic HDD?

2. Why use a different password for each account?
   A) Make all passwords shorter
   B) Remove the need for login
   C) Give everyone equal access
   D) Limit damage after a leak

3. How does encryption protect transmitted data?

4. A 20 x 10 image at 8 bits per pixel has 200 bytes of raw pixel data, but its uncompressed file is larger. Explain why this need not be an error and name two metadata examples.

Today's objectives

Identify ethical, cultural, environmental and privacy effects of technology. Explain who benefits or could be disadvantaged.

Key vocabulary - Read, cover, recall

Ethics
Judgements about what is fair, right or responsible.

Digital divide
Unequal access to digital devices, connectivity and skills.

Privacy
Control and protection of information about individuals.

E-waste
Discarded electrical or electronic equipment.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Ethics versus legislation

Ethical questions concern what is fair or responsible, even when an action is legal. Legal issues concern laws that apply. Consider the effect on different groups, not only technical capability.

Society and culture

Digital services make communication and collaboration easier, but unequal access to devices or reliable Internet creates a digital divide. Global digital communication can also affect language, culture and social interaction.

Environment

Electronic devices require energy and materials to produce, run and dispose of. Repair and recycling can reduce e-waste; increased energy demand has environmental costs.

Privacy and fair decisions

Data collection may improve services but can expose personal information. Biased automated decisions or inaccessible technology can disadvantage people, so design should be fair, transparent and inclusive.

I Do - Worked model

A town library introduces an online-only booking service. Explain TWO ethical or social concerns that the library should consider. [4 marks]

- Identify the users: some visitors may have no broadband or suitable device.
- Online-only booking could exclude those without digital access [1], preventing them from booking services easily [1].
- Some visitors may lack digital skills or accessible interfaces [1], creating barriers to fair use [1].
- Two concerns and linked effects. The library could also provide an in-person/telephone option.

We Do - Guided practice

1. A health centre makes appointments mainly through a mobile app. Explain two ways this could affect different patients. [4 marks]

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2. A school replaces working tablets every two years. Explain an environmental concern and a possible mitigation. [4 marks]

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3. An online service collects location data to recommend local events. Explain one benefit and one privacy risk. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the access alternative displayed.

```
hasInternet = false
if hasInternet then
print("Online form")
else
print("Paper form")
endif
```

You Do - Independent exam questions

Q2. State TWO environmental concerns linked to digital technology. [2 marks]

Q3. A secondary school moves all homework online. Explain one benefit and one social disadvantage. [4 marks]

Q4. Evaluate how a council's move towards digital-only public services could affect citizens and the environment. [6 marks]

Exit Ticket - True or False?

A service can be legal but still raise ethical concerns about fairness and access.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It brings file fragments together, reducing read-head movement when accessing a file.

2. D - Limit damage after a leak

3. Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key.

4. The file may include metadata or a header as well as pixels. Width, height and colour depth are examples. The formula counts only pixel data.

I Do - Explain the model

Identify the users: some visitors may have no broadband or suitable device.

Online-only booking could exclude those without digital access [1], preventing them from booking services easily [1].

Some visitors may lack digital skills or accessible interfaces [1], creating barriers to fair use [1].

Two concerns and linked effects. The library could also provide an in-person/telephone option.

We Do - Model answers

1. Patients who have smartphones can book appointments remotely [1], making access more convenient [1]. Patients without devices or with accessibility difficulties may be excluded [1], making healthcare harder to access [1].

2. Frequent replacement increases e-waste/material demand [1], creating disposal and manufacturing impacts [1]. Repair, reuse or recycle functioning tablets [1], reducing waste and demand for new materials [1].

3. Location data can enable more relevant nearby recommendations [1], helping users find suitable events [1]. Storing precise location histories [1] risks revealing where users visit if data is misused or exposed [1].

Code First - Answer

```
hasInternet = false
if hasInternet then
print("Online form")
else
print("Paper form")
endif
```

Expected output: Paper form

You Do - Self-mark

Q2: E-waste from discarded devices [1] and electricity/energy consumption [1]. Other clear valid concerns include manufacturing emissions or extracting raw materials.

Q3: Homework can be accessed anywhere with Internet [1], making resources easy to distribute/submit [1]. Pupils without reliable devices or connectivity [1] may find it harder to complete work, widening inequality [1].

Q4: Digital services can reduce travel/paper use [1], improving convenience and possibly resource use [1]. They may exclude people without reliable devices/skills [1], creating an access barrier [1]. Data collection could compromise privacy [1] without suitable protections, so a justified balance including alternative access is important [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.